

DAY 16 — WIRESHARK REPORT

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1. What is Wireshark?

Wireshark is a network protocol analyzer. It captures and displays packets so users can inspect how devices communicate, troubleshoot network issues, and investigate authorized security incidents.

2. What is packet capture?

Packet capture is the process of recording network packets travelling through a selected interface. The captured data can later be examined to understand communication, protocols, timing and errors. It should only be done on traffic that I am authorized to inspect.

3. What is a capture interface?

A capture interface is the network adapter or interface from which Wireshark collects packets, such as Wi-Fi, Ethernet, or a loopback interface. Choosing the correct interface determines which traffic can be observed.

4. Name the three main Wireshark panes.

The three main panes are the Packet List pane, Packet Details pane, and Packet Bytes pane. They show a summary of packets, the selected packet's decoded protocol fields, and the packet's raw bytes respectively.

5. What is the difference between source and destination IP?

The source IP identifies the IP address from which a packet is sent, while the destination IP identifies the address to which it is being sent. These values describe the direction of that packet at the IP layer.

6. What does the Protocol column show?

The Protocol column shows the protocol Wireshark identifies for a packet, such as TCP, UDP, DNS, ICMP or TLS. It helps quickly classify the type of network communication.

7. What is a Wireshark display filter?

A display filter is an expression used to show only packets matching specified conditions in an existing capture or live view. It does not delete other packets from the capture. For example, `dns` displays packets Wireshark recognizes as DNS.

8. Which filter shows ICMP traffic?

Use the display filter ``icmp`` to show IPv4 ICMP packets. For IPv6 ICMP, use ``icmpv6``.

9. Which filter shows DNS traffic?

Use ``dns`` as the display filter to show packets decoded as DNS.

10. Which filter shows TCP traffic?

Use ``tcp`` to display TCP packets.

11. What does `tcp.port == 443` display?

The filter ``tcp.port == 443`` displays TCP packets where either the source or destination TCP port is 443. Port 443 is commonly used for HTTPS, although the port number alone does not prove the application data is HTTPS.

12. What is DNS?

DNS stands for Domain Name System. It is a distributed naming system that helps translate domain names, such as `example.com`, into IP addresses and provides other records used by internet services.

13. What is an ICMP Echo Request?

An ICMP Echo Request is a message commonly sent by the ping utility to ask another host whether it is reachable. It includes an identifier and sequence information that help match the response.

14. What is an ICMP Echo Reply?

An ICMP Echo Reply is the response to an Echo Request. It indicates that the destination host or network stack received and answered the request, although it does not guarantee that every service on that host is available.

15. What is the TCP three-way handshake?

The TCP three-way handshake is the process used to establish a TCP connection. The initiating side sends SYN, the receiving side responds with SYN-ACK, and the initiator completes the setup with ACK. After this exchange, data can be transmitted.

16. What are SYN, SYN-ACK and ACK?

SYN is a TCP flag used to initiate a connection and synchronize sequence numbers. SYN-ACK acknowledges the SYN and sends the responder's own SYN. ACK acknowledges received data or handshake information. Together, these flags form the normal TCP connection setup.

17. Why is HTTPS content normally not visible as plain text?

HTTPS protects HTTP communication using TLS encryption. A packet capture can often show connection metadata, such as IP addresses, ports and some handshake information, but the application content is encrypted and cannot normally be read as plain text without authorized decryption material and appropriate configuration.

18. What is a PCAP/PCAPNG file?

PCAP and PCAPNG are file formats used to store captured network packets for later analysis. PCAPNG is a newer, extensible format that can store additional capture information, such as interface details and metadata.

19. Why is Wireshark useful in incident response?

Wireshark helps incident responders examine authorized packet captures to understand communication patterns, identify unexpected connections, review protocol behavior, and establish a timeline of network activity. It provides evidence that can support investigation alongside logs and endpoint data.

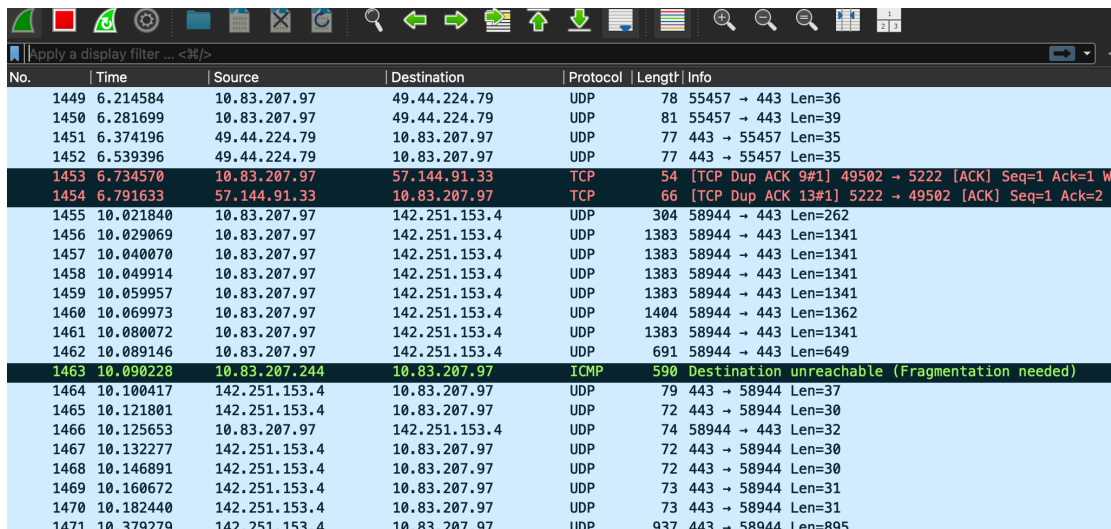
20. Why must packet capture be performed only on authorized traffic?

Network packets may contain personal information, private messages, credentials or other sensitive data. Capturing traffic without permission can violate privacy, organizational policy or law. I should capture only my own lab

traffic or traffic for which I have explicit authorization, and review screenshots and files before sharing them.

Screenshots:-

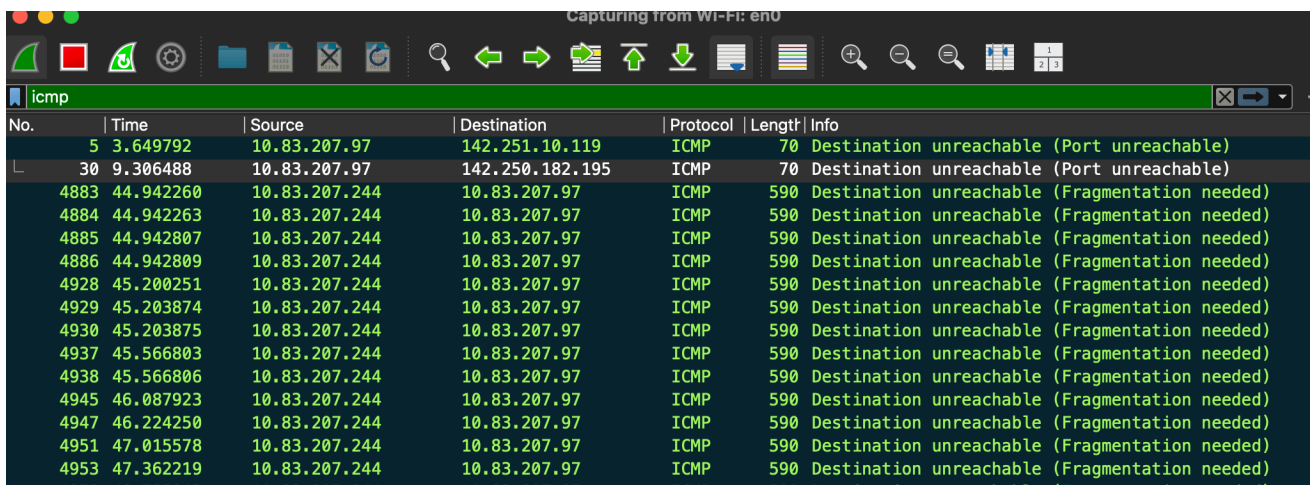
- Wireshark interface showing the selected authorized capture interface.



The screenshot shows the Wireshark interface with a display filter applied. The packet list pane shows a series of packets, with the selected packet (No. 1463) highlighted. The packet details pane shows the selected packet's structure, including the ICMP header and the message type (Destination unreachable).

No.	Time	Source	Destination	Protocol	Length	Info
1449	6.214584	10.83.207.97	49.44.224.79	UDP	78	55457 → 443 Len=36
1450	6.281699	10.83.207.97	49.44.224.79	UDP	81	55457 → 443 Len=39
1451	6.374196	49.44.224.79	10.83.207.97	UDP	77	443 → 55457 Len=35
1452	6.539396	49.44.224.79	10.83.207.97	UDP	77	443 → 55457 Len=35
1453	6.734570	10.83.207.97	57.144.91.33	TCP	54	[TCP Dup ACK 9#1] 49502 → 5222 [ACK] Seq=1 Ack=1 W
1454	6.791633	57.144.91.33	10.83.207.97	TCP	66	[TCP Dup ACK 13#1] 5222 → 49502 [ACK] Seq=1 Ack=2
1455	10.021840	10.83.207.97	142.251.153.4	UDP	304	58944 → 443 Len=262
1456	10.029069	10.83.207.97	142.251.153.4	UDP	1383	58944 → 443 Len=1341
1457	10.040070	10.83.207.97	142.251.153.4	UDP	1383	58944 → 443 Len=1341
1458	10.049914	10.83.207.97	142.251.153.4	UDP	1383	58944 → 443 Len=1341
1459	10.059957	10.83.207.97	142.251.153.4	UDP	1383	58944 → 443 Len=1341
1460	10.069973	10.83.207.97	142.251.153.4	UDP	1404	58944 → 443 Len=1362
1461	10.080072	10.83.207.97	142.251.153.4	UDP	1383	58944 → 443 Len=1341
1462	10.089146	10.83.207.97	142.251.153.4	UDP	691	58944 → 443 Len=649
1463	10.090228	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
1464	10.100417	142.251.153.4	10.83.207.97	UDP	79	443 → 58944 Len=37
1465	10.121801	142.251.153.4	10.83.207.97	UDP	72	443 → 58944 Len=30
1466	10.125653	10.83.207.97	142.251.153.4	UDP	74	58944 → 443 Len=32
1467	10.132277	142.251.153.4	10.83.207.97	UDP	72	443 → 58944 Len=30
1468	10.146891	142.251.153.4	10.83.207.97	UDP	72	443 → 58944 Len=30
1469	10.160672	142.251.153.4	10.83.207.97	UDP	73	443 → 58944 Len=31
1470	10.182440	142.251.153.4	10.83.207.97	UDP	73	443 → 58944 Len=31
1471	10.370279	142.251.153.4	10.83.207.97	UDP	937	443 → 58944 Len=895

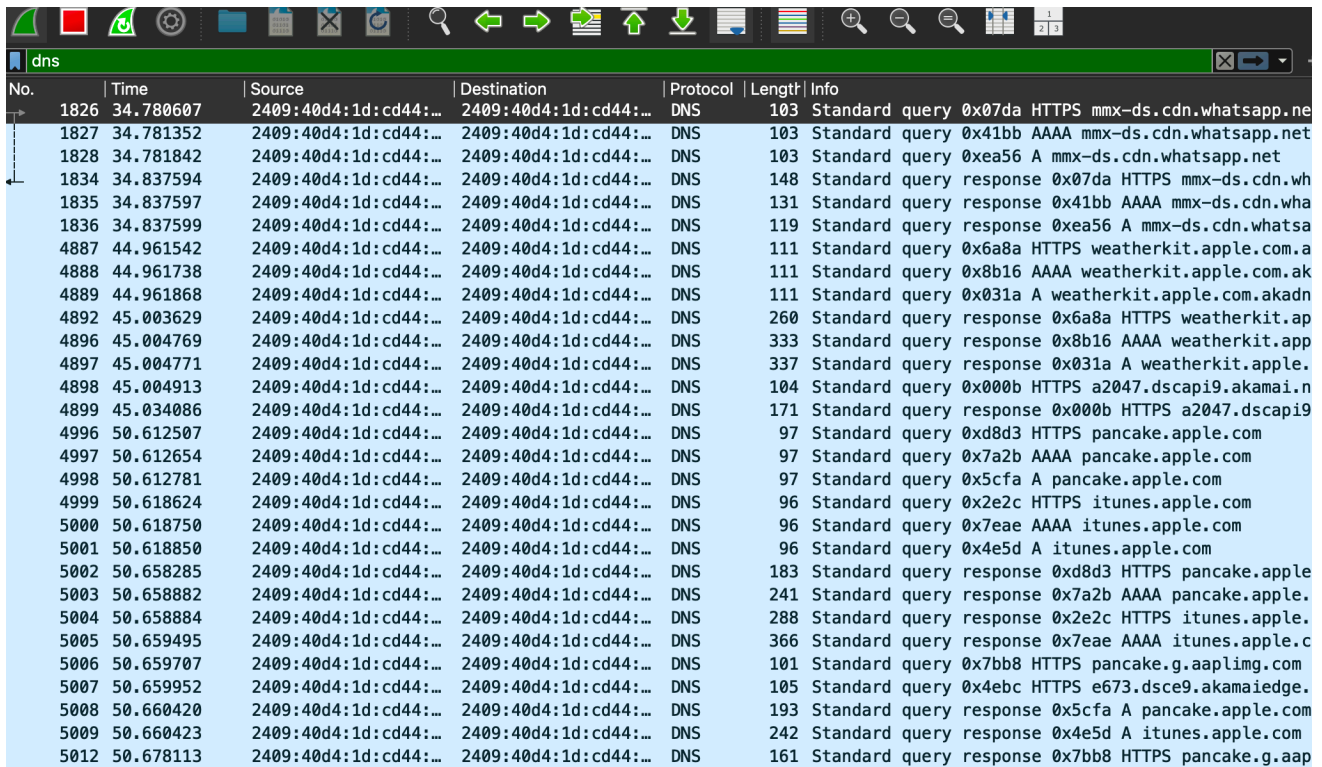
- Display filter `icmp` applied to your own lab capture.



The screenshot shows the Wireshark interface with a display filter applied. The packet list pane shows a series of packets, with the selected packet (No. 5) highlighted. The packet details pane shows the selected packet's structure, including the ICMP header and the message type (Destination unreachable).

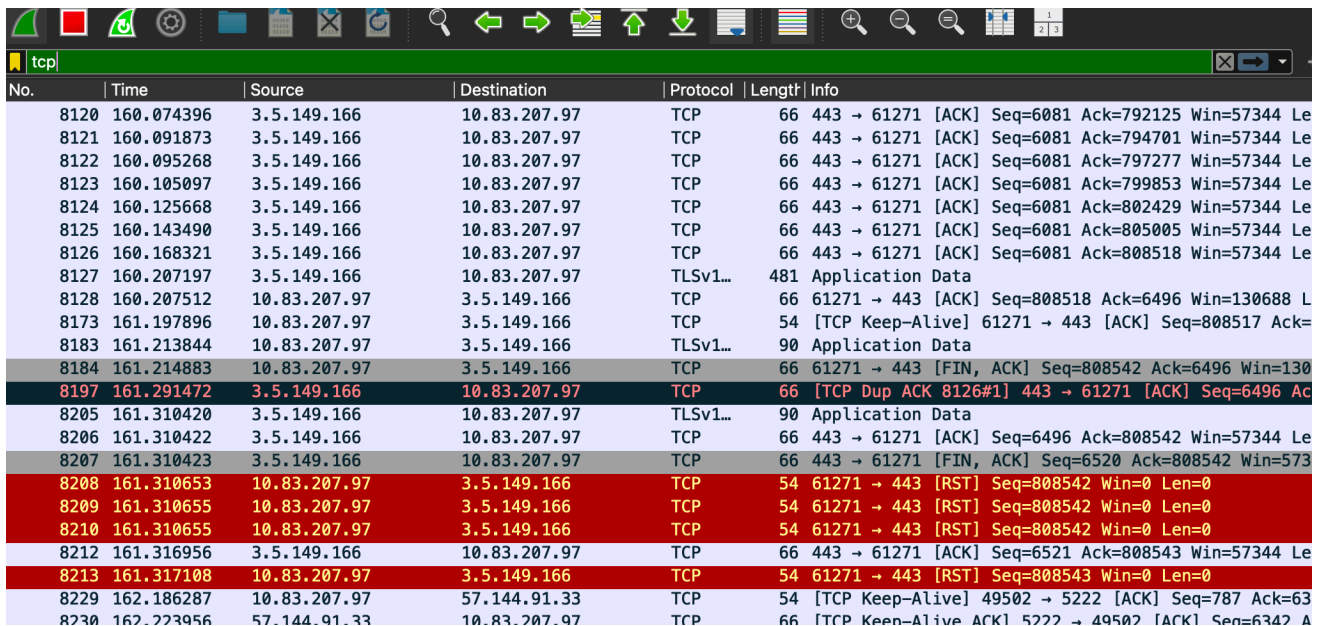
No.	Time	Source	Destination	Protocol	Length	Info
5	3.649792	10.83.207.97	142.251.10.119	ICMP	70	Destination unreachable (Port unreachable)
30	9.306488	10.83.207.97	142.250.182.195	ICMP	70	Destination unreachable (Port unreachable)
4883	44.942260	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4884	44.942263	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4885	44.942807	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4886	44.942809	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4928	45.200251	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4929	45.203874	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4930	45.203875	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4937	45.566803	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4938	45.566806	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4945	46.087923	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4947	46.224250	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4951	47.015578	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)
4953	47.362219	10.83.207.244	10.83.207.97	ICMP	590	Destination unreachable (Fragmentation needed)

- Display filter `dns` applied to your own lab capture.



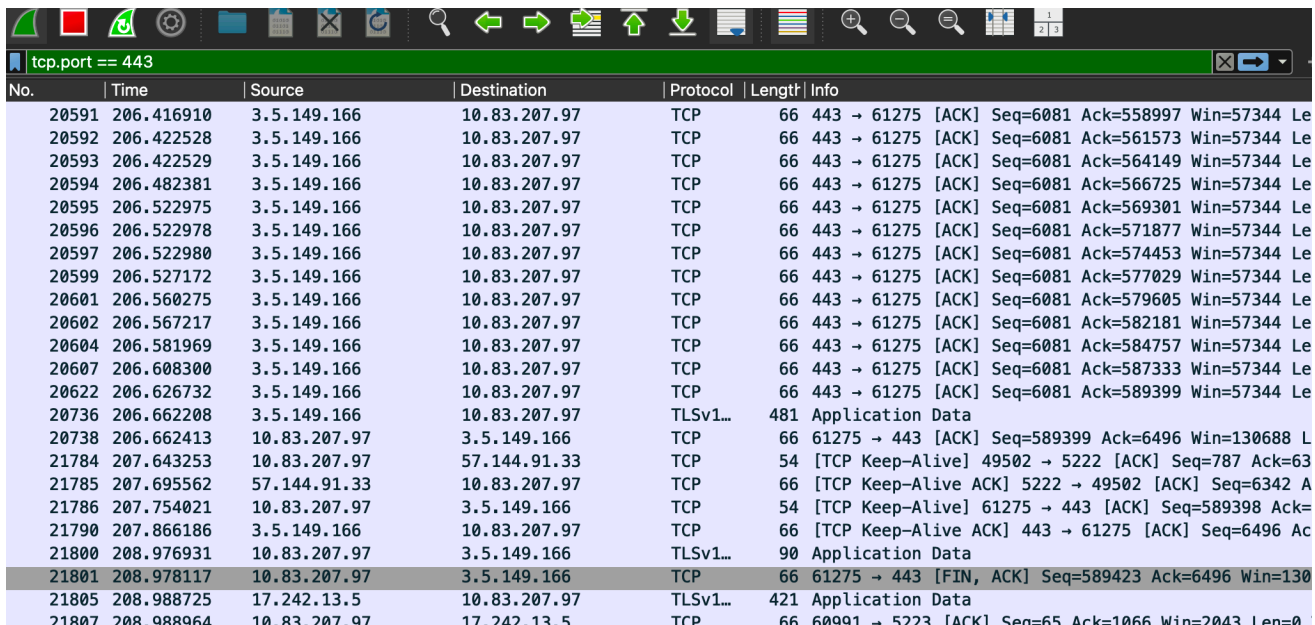
No.	Time	Source	Destination	Protocol	Length	Info
1826	34.780607	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	103	Standard query 0x07da HTTPS mmx-ds.cdn.whatsapp.net
1827	34.781352	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	103	Standard query 0x41bb AAAA mmx-ds.cdn.whatsapp.net
1828	34.781842	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	103	Standard query 0xea56 A mmx-ds.cdn.whatsapp.net
1834	34.837594	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	148	Standard query response 0x07da HTTPS mmx-ds.cdn.whatsapp.net
1835	34.837597	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	131	Standard query response 0x41bb AAAA mmx-ds.cdn.whatsapp.net
1836	34.837599	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	119	Standard query response 0xea56 A mmx-ds.cdn.whatsapp.net
4887	44.961542	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	111	Standard query 0x6a8a HTTPS weatherkit.apple.com.akadn
4888	44.961738	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	111	Standard query 0x8b16 AAAA weatherkit.apple.com.akadn
4889	44.961868	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	111	Standard query 0x031a A weatherkit.apple.com.akadn
4892	45.003629	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	260	Standard query response 0x6a8a HTTPS weatherkit.apple.com
4896	45.004769	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	333	Standard query response 0x8b16 AAAA weatherkit.apple.com
4897	45.004771	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	337	Standard query response 0x031a A weatherkit.apple.com
4898	45.004913	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	104	Standard query 0x000b HTTPS a2047.dscapi9.akamai.net
4899	45.034086	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	171	Standard query response 0x000b HTTPS a2047.dscapi9.akamai.net
4996	50.612507	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	97	Standard query 0xd8d3 HTTPS pancake.apple.com
4997	50.612654	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	97	Standard query 0x7a2b AAAA pancake.apple.com
4998	50.612781	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	97	Standard query 0x5cfa A pancake.apple.com
4999	50.618624	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	96	Standard query 0x2e2c HTTPS itunes.apple.com
5000	50.618750	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	96	Standard query 0x7eae AAAA itunes.apple.com
5001	50.618850	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	96	Standard query 0x4e5d A itunes.apple.com
5002	50.658285	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	183	Standard query response 0xd8d3 HTTPS pancake.apple.com
5003	50.658882	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	241	Standard query response 0x7a2b AAAA pancake.apple.com
5004	50.658884	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	288	Standard query response 0x2e2c HTTPS itunes.apple.com
5005	50.659495	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	366	Standard query response 0x7eae AAAA itunes.apple.com
5006	50.659707	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	101	Standard query 0x7bb8 HTTPS pancake.g.aaplimg.com
5007	50.659952	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	105	Standard query 0x4ebc HTTPS e673.ds9.akamaiedge.net
5008	50.660420	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	193	Standard query response 0x5cfa A pancake.apple.com
5009	50.660423	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	242	Standard query response 0x4e5d A itunes.apple.com
5012	50.678113	2409:40d4:1d:cd44::...	2409:40d4:1d:cd44::...	DNS	161	Standard query response 0x7bb8 HTTPS pancake.g.aaplimg.com

- Display filter `tcp` applied to your own lab capture.



No.	Time	Source	Destination	Protocol	Length	Info
8120	160.074396	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=792125 Win=57344 Len=0
8121	160.091873	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=794701 Win=57344 Len=0
8122	160.095268	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=797277 Win=57344 Len=0
8123	160.105097	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=799853 Win=57344 Len=0
8124	160.125668	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=802429 Win=57344 Len=0
8125	160.143490	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=805005 Win=57344 Len=0
8126	160.168321	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6081 Ack=808518 Win=57344 Len=0
8127	160.207197	3.5.149.166	10.83.207.97	TLSv1...	481	Application Data
8128	160.207512	10.83.207.97	3.5.149.166	TCP	66	61271 → 443 [ACK] Seq=808518 Ack=6496 Win=130688 Len=0
8173	161.197896	10.83.207.97	3.5.149.166	TCP	54	[TCP Keep-Alive] 61271 → 443 [ACK] Seq=808517 Ack=6496
8183	161.213844	10.83.207.97	3.5.149.166	TLSv1...	90	Application Data
8184	161.214883	10.83.207.97	3.5.149.166	TCP	66	61271 → 443 [FIN, ACK] Seq=808542 Ack=6496 Win=130688 Len=0
8197	161.291472	3.5.149.166	10.83.207.97	TCP	66	[TCP Dup ACK 8126#1] 443 → 61271 [ACK] Seq=6496 Ack=808542
8205	161.310420	3.5.149.166	10.83.207.97	TLSv1...	90	Application Data
8206	161.310422	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6496 Ack=808542 Win=57344 Len=0
8207	161.310423	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [FIN, ACK] Seq=6520 Ack=808542 Win=57344 Len=0
8208	161.310653	10.83.207.97	3.5.149.166	TCP	54	61271 → 443 [RST] Seq=808542 Win=0 Len=0
8209	161.310655	10.83.207.97	3.5.149.166	TCP	54	61271 → 443 [RST] Seq=808542 Win=0 Len=0
8210	161.310655	10.83.207.97	3.5.149.166	TCP	54	61271 → 443 [RST] Seq=808542 Win=0 Len=0
8212	161.316956	3.5.149.166	10.83.207.97	TCP	66	443 → 61271 [ACK] Seq=6521 Ack=808543 Win=57344 Len=0
8213	161.317108	10.83.207.97	3.5.149.166	TCP	54	61271 → 443 [RST] Seq=808543 Win=0 Len=0
8229	162.186287	10.83.207.97	57.144.91.33	TCP	54	[TCP Keep-Alive] 49502 → 5222 [ACK] Seq=787 Ack=63
8230	162.223956	57.144.91.33	10.83.207.97	TCP	66	[TCP Keep-Alive ACK] 5222 → 49502 [ACK] Seq=6342 Ack=787

- Display filter `tcp.port == 443` applied to your own lab capture (if your authorized lab generates matching traffic).



The image shows a Wireshark packet capture window with the display filter 'tcp.port == 443' applied. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
20591	206.416910	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=558997 Win=57344 Le
20592	206.422528	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=561573 Win=57344 Le
20593	206.422529	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=564149 Win=57344 Le
20594	206.482381	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=566725 Win=57344 Le
20595	206.522975	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=569301 Win=57344 Le
20596	206.522978	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=571877 Win=57344 Le
20597	206.522980	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=574453 Win=57344 Le
20599	206.527172	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=577029 Win=57344 Le
20601	206.560275	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=579605 Win=57344 Le
20602	206.567217	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=582181 Win=57344 Le
20604	206.581969	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=584757 Win=57344 Le
20607	206.608300	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=587333 Win=57344 Le
20622	206.626732	3.5.149.166	10.83.207.97	TCP	66	443 → 61275 [ACK] Seq=6081 Ack=589399 Win=57344 Le
20736	206.662208	3.5.149.166	10.83.207.97	TLSv1...	481	Application Data
20738	206.662413	10.83.207.97	3.5.149.166	TCP	66	61275 → 443 [ACK] Seq=589399 Ack=6496 Win=130688 L
21784	207.643253	10.83.207.97	57.144.91.33	TCP	54	[TCP Keep-Alive] 49502 → 5222 [ACK] Seq=787 Ack=63
21785	207.695562	57.144.91.33	10.83.207.97	TCP	66	[TCP Keep-Alive ACK] 5222 → 49502 [ACK] Seq=6342 A
21786	207.754021	10.83.207.97	3.5.149.166	TCP	54	[TCP Keep-Alive] 61275 → 443 [ACK] Seq=589398 Ack=
21790	207.866186	3.5.149.166	10.83.207.97	TCP	66	[TCP Keep-Alive ACK] 443 → 61275 [ACK] Seq=6496 Ac
21800	208.976931	10.83.207.97	3.5.149.166	TLSv1...	90	Application Data
21801	208.978117	10.83.207.97	3.5.149.166	TCP	66	61275 → 443 [FIN, ACK] Seq=589423 Ack=6496 Win=130
21805	208.988725	17.242.13.5	10.83.207.97	TLSv1...	421	Application Data
21807	208.988964	10.83.207.97	17.242.13.5	TCP	66	60991 → 5223 [ACK] Seq=65 Ack=1066 Win=2043 Len=0

Conclusion

This assignment helped me understand how Wireshark organizes packet information and how display filters make analysis more focused. I learned the purpose of common protocols, the TCP handshake, DNS, ICMP and encrypted HTTPS traffic. I also learned that packet analysis must be performed responsibly, using only authorized traffic and protecting sensitive information.